Practice workbook · Describe what red cells and plasma carry, and from where to where

Function of red blood cells and plasma in carrying oxygen, nutrients, wastes, hormones, heat and drugs around the body.

6 question parts: 1 worked-example retry and 5 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DESCRIBE-BLOOD-TRANSPORT-ROLES

1. Question 6 · part b · 5 marks

June 2024 Paper 1

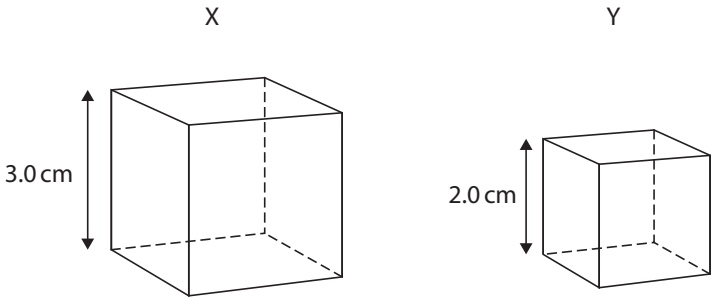
Worked-example retry · Attempt part b only.

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6 (a) Humans have a small surface area to volume ratio (SA : V) compared with some other organisms.

Cube X represents a human.

Cube Y represents an organism with a larger SA : V.



The table gives the SA : V of cube Y.

Cube	Surface area in cm^2	Volume in cm^3	SA : V
X			
Y	24.0	8.0	3:1

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(i) Calculate the surface area of cube X.

(2)

surface area = ... cm^2

(ii) Calculate the volume of cube X.

(2)

volume = ... cm^3

(iii) Calculate the SA : V of cube X.

(2)

SA : V = ... : 1

1. Question 6 · part b · 5 marks

June 2024 Paper 1

Worked-example retry · Attempt part b only.

(b) Multicellular organisms, with a small SA:V ratio, such as humans, need an efficient transport system to deliver substances to body cells.

Describe the role of the human blood transport system in delivering substances to body cells.

(5)

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1. Question 6 · part b · 5 marks

June 2024 Paper 1

Worked-example retry · Attempt part b only.

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(c) Bacteria are able to exchange materials directly with their environment as they have a very high SA : V ratio.

They do not have an internal transport system.

Describe how bacteria exchange materials with their environment.

(2)

(Total for Question 6 = 13 marks)

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1. Question 6 · part b · 5 marks · Working and answer

Working and answer

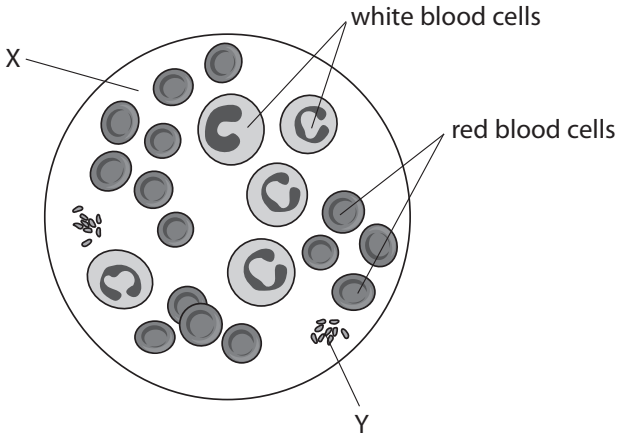
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2. Question 1 · part a · 1 mark

November 2020 Paper 1R

Further practice · Attempt part a only.

1 The diagram shows a sample of human blood.



(a) State the function of red blood cells.

(1)

.....

.....

(b) Give the name of X and Y.

(2)

X .....

Y .....

(c) (i) Calculate the ratio of red blood cells to white blood cells.

Write the ratio in its simplest form.

(2)

ratio = ...

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2. Question 1 · part a · 1 mark · Working and answer

Working and answer

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3. Question 1 · part a.i · 3 marks

November 2021 Paper 2

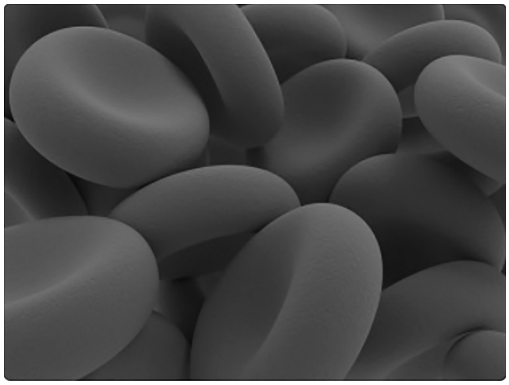
Further practice · Attempt part a.i only.

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1 (a) The photograph shows a group of red blood cells.



(Source: RomanenkoAlexey/Shutterstock)

(i) Describe the function of red blood cells.

(3)

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(ii) Explain how the structure of the red blood cell is adapted to its function.

(4)

3. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

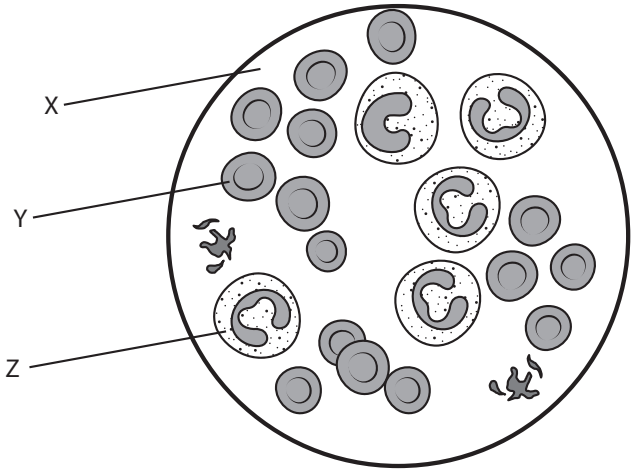
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4. Question 2 · part a.ii · 2 marks

June 2024 Paper 1R

Further practice · Attempt part a.ii only.

2 (a) The diagram shows components, X, Y and Z, in a sample of human blood.



(i) Name components X, Y and Z.

(3)

X .....

Y .....

Z .....

(ii) Describe the function of component Y.

(2)

.....

.....

.....

.....

(iii) A sample of blood from a patient is analysed in a hospital.

A scientist finds that the number of component Z in the blood sample is higher than normal.

State a possible reason for the increase in the number of component Z.

(1)

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4. Question 2 · part a.ii · 2 marks · Working and answer

Working and answer

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5. Question 5 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

5 A scientist investigates how the drug, atropine, affects the heart rate in humans.

Atropine paralyses the vagus nerve which carries impulses from the brain to the heart.

The heart rates of ten people are measured at intervals of two minutes for eight minutes, after which, atropine is injected.

The scientist continues to measure the heart rates.

The scientist's results are shown in the table.

Time in minutes	Heart rate in beats per minute
0	90
2	81
4	78
6	80
8	82
atropine injected → 9	100
10	120
11	148
12	160
13	138
15	128
17	120
19	116
21	109
23	100

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5. Question 5 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

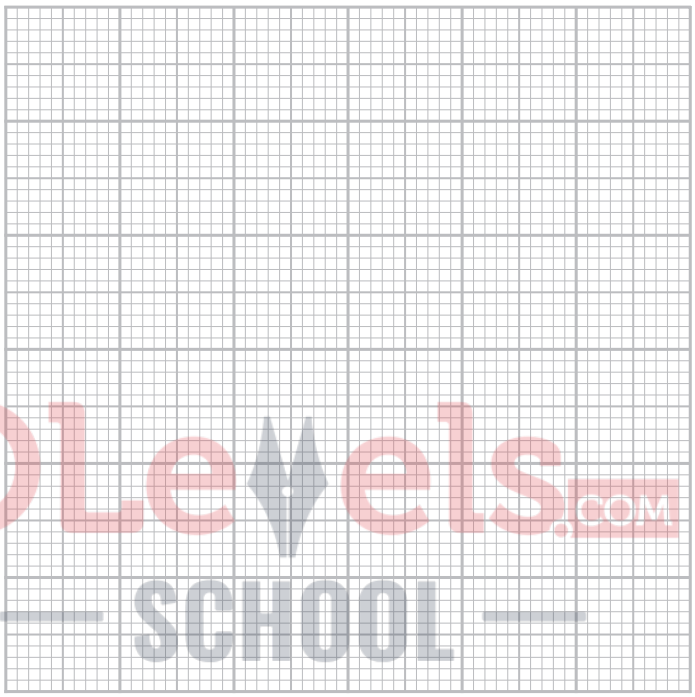
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(a) (i) Plot the scientist's results on the grid. Draw a curve of best fit.

(6)



(ii) Explain why ten people are injected with atropine rather than just one person.

(2)

(b) (i) Calculate the percentage change in the heart rate at 12 minutes compared to the heart rate at the start of the investigation.

(2)

percentage change = ...%

5. Question 5 · part b.ii · 2 marks

November 2020 Paper 2R

Further practice · Attempt part b.ii only.

(ii) Explain why this percentage change is not achieved as soon as atropine is injected.

(2)

(c) Suggest the function of the vagus nerve.

(2)

(Total for Question 5 = 14 marks)

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Working and answer

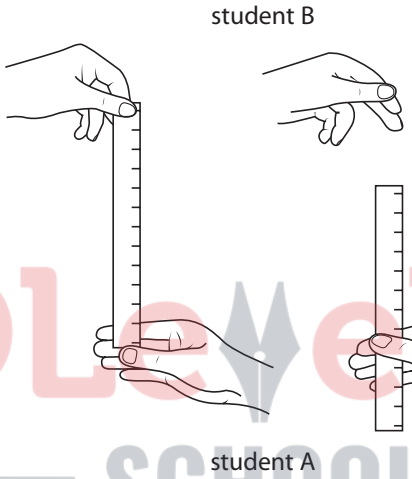
6. Question 5 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

5 Two students use this method to investigate if caffeine affects the speed of a reflex action.

- student A sits in a chair holding out a hand with their thumb and forefinger open
- student B holds a metre rule with the 0 mm mark just above the open thumb and forefinger of student A
- student B releases the ruler
- student A catches the ruler as quickly as possible by closing their thumb and forefinger



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The method is repeated 10 times.

The student catching the ruler now drinks a cup of coffee and waits for 30 minutes.

The investigation is then repeated.

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6. Question 5 · part c · 3 marks

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Further practice · Attempt part c only.

The table shows the student's results.

Conditions \ Trial	Distance travelled by ruler in mm										
	1	2	3	4	5	6	7	8	9	10	mean
without coffee	210	190	200	205	207	195	198	195	199	201	200
with coffee	165	160	163	167	165	170	175	170	180	175	

(a) Complete the table by calculating the missing mean value.

(2)

(b) Coffee contains caffeine.

State the effect of caffeine on the reflex action.

(1)

(c) Explain why the student waited 30 minutes after drinking the coffee before continuing with the investigation.

(3)

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6. Question 5 · part c · 3 marks

November 2021 Paper 2

Further practice · Attempt part c only.

(d) The student who catches the ruler does not eat or drink anything for several hours before the investigation.

Suggest why this improves the validity of the investigation.

(2)

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(e) A cup of coffee contains 100 mg of caffeine. A can of cola contains 50 mg of caffeine.

If the student catching the ruler had drunk a can of cola instead of coffee, estimate the mean distance travelled by the ruler.

(1)

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mean distance = ... mm

(Total for Question 5 = 9 marks)

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6. Question 5 · part c · 3 marks · Working and answer

Working and answer

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